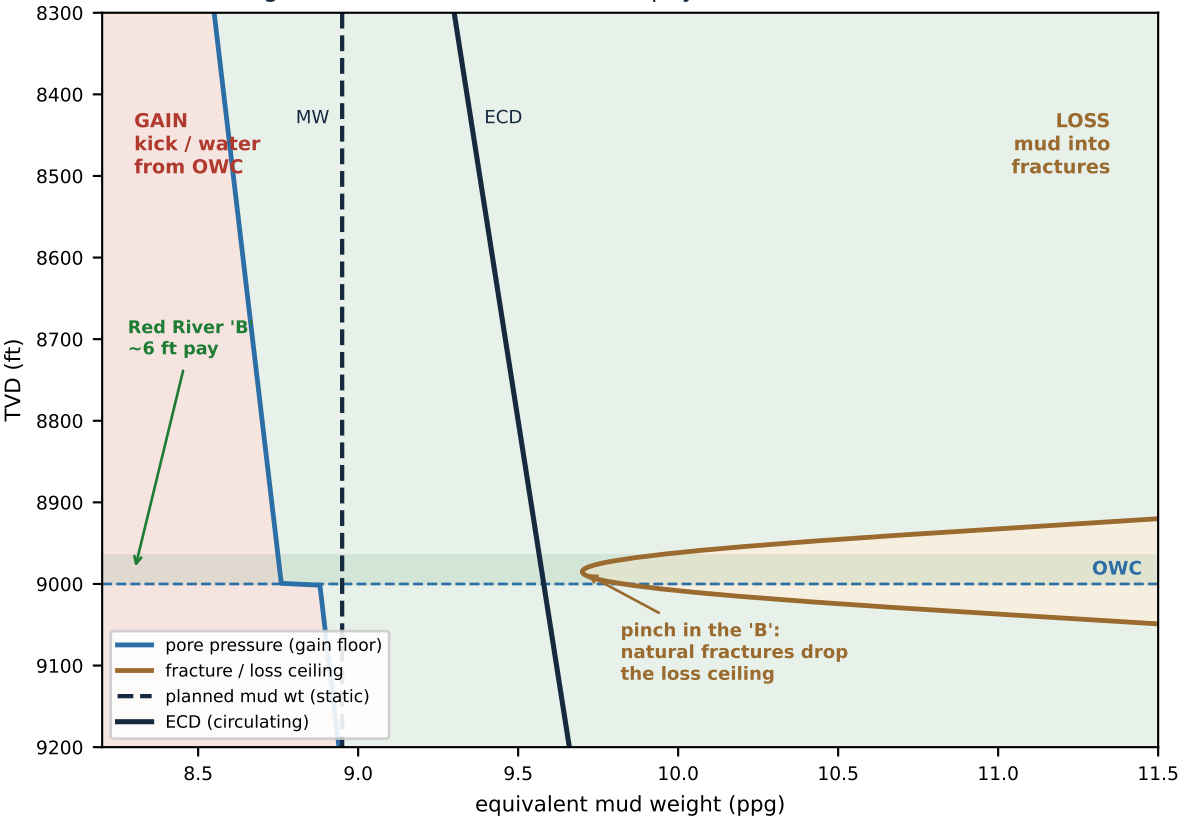


Fluid Balance — Loss & Gain Risk

Drilling-system fluid (mud / ECD) vs formation pressure — the mud-weight window pinches inside the pay, where losses and gains meet

The mud-weight window narrows inside the pay



TWO WAYS THE FLUID BALANCE FAILS

OVERBALANCED — ECD above formation pressure

→ LOSSES: mud invades the natural fractures; you lose returns, burn mud, and can drill blind to a kick building below.

UNDERBALANCED — ECD below formation pressure

← GAIN / KICK: formation fluid enters the well; here that means WATER pulled up from the OWC sitting just under the 'B'.

LOSS / GAIN RISK BY PHASE

phase	LOSS	GAIN
Curve	Low	Low
Landing	Med	Med
Lateral	High	Med
Trips / connections	Med	Med

LOSS MANAGEMENT

- LCM sized to the fractures, staged & ready
- hold ECD down: flow rate, hole cleaning, RPM, slow connections & trips (surge)
- cure losses before drilling ahead — don't chase
- air-mist / UBD only where clear of the OWC

GAIN / WELL CONTROL

- flow checks + active kick detection
- trip & connection discipline (avoid swab)
- keep a trip margin; PWD to watch BHP live
- geosteer to hold standoff ABOVE the OWC — the cheapest gain insurance there is

NARROW-WINDOW TOOL · MPD

Managed-pressure drilling: closed-loop, holds a precise bottomhole pressure and reacts in seconds to BOTH losses and gains — the right tool where the loss ceiling and gain floor pinch in the pay. With the OWC right below, hold a precise slight overbalance.